

NATIONAL OCCUPATIONAL STANDARD FOR MAINTENANCE TECHNICIAN



**NOS.MT.01
FIRST EDITION**

APPROVING AUTHORITY

This National Occupational Standard (NOS) has been prepared and published under the authority of the Zambia Qualifications Authority Board on 15th February, 2024.

ZAMBIA QUALIFICATIONS AUTHORITY

The Zambia Qualifications Authority (ZAQA) Act No. 13 of 2011 was enacted by the Government of the Republic of Zambia to ***“provide for the development and implementation of a national qualifications framework; establish the Zambia Qualifications Authority; provide for the registration and accreditation of qualifications; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing”***. Among other functions, ZAQA is responsible for ***determining national standards for any occupation***, through the various sector specific National Occupational Standards Development Teams (NOSDTs).

REVISION OF NATIONAL OCCUPATIONAL STANDARDS

National Occupational Standards (NOS) shall be revised every 5 years, or whenever it is deemed necessary, by the issuance of either amendments or of revised editions. It is important that users of the NOS ascertain that they are in possession of the latest amendments or editions.

NOS DEVELOPMENT TEAM RESPONSIBLE

This National Occupational Standard was prepared by the Manufacturing NOSDT, upon which the following organisations were represented:

1. Biomedical Society of Zambia (BMSZ)
2. Copperbelt University (CBU)
3. Engineering Institution of Zambia (EIZ)
4. Good Time Steel Company Zambia Limited (GTS)
5. Lusaka Vocational and Technical College (LVTC)
6. Makeni Ecumenical Centre (MEC)
7. Ministry of Commerce, Trade and Industry (MCTI)
8. Ministry of Health (MoH)
9. National Institute for Scientific and Industrial Research (NISIR)
10. Northern Technical College (NORTEC)
11. University of Lusaka (UNILUS)
12. University of Zambia (UNZA)
13. Zambia Association of Manufacturers (ZAM)
14. Zambia Forestry College (ZFC)
15. Zambia Qualifications Authority (ZAQA) – Secretariat

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6. Mr. Patrice Shawa (Zambia Forestry College)
7. Mr. Pascal Sikainga (Makeni Ecumenical Centre)
8. Mr Moses Ngosa (Ministry of Commerce, Trade and Industry)
9. Eng. Mark Kapasa Mutimushi (Northern Technical College)
10. Dr. James Phiri (University of Zambia)
11. Mr. Durban K Kambaki (Good Time Steel Company Zambia Limited)
12. Mr. Joseph Banda - Individual Expert
13. Mr. Mutale Christopher (Lusaka Vocational and Technical College)
14. Dr. Diana Kangwa (Engineering Institution of Zambia)
15. Mr. Fidelis Cheelo (Zambia Qualifications Authority)
16. Mr. Jericho Kashiya (Zambia Qualifications Authority)
17. Ms. Womba Soneka (Zambia Qualifications Authority)

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FOREWORD

The Zambia Qualifications Authority (ZAQA) is a statutory body under the Ministry of Education established by ZAQA Act No. 13 of 2011 to ***“develop and implement a national qualifications framework; register and accredit qualifications; and ensure that standards and registered qualifications are internationally comparable.”***

Among other functions, ZAQA is responsible for ***“determining national standards for any occupation”***, through the various sector specific National Occupational Standards Development Teams (NOSDTs) with experts composed of representation from the appropriate authorities, government departments, consumer associations, regulators, industry, academia, and non-governmental organisations, etc.

This National Occupational Standard (NOS) has been developed by the Manufacturing National Occupational Standards Development Team in accordance with the laid down procedures and guidelines of ZAQA. All users should ensure that they have the latest edition of this publication as the National Occupational Standards are revised from time to time.

This NOS shall be used by, among others, industry, employers, quality assurance bodies, awarding and professional bodies, and education and training institutions, as a benchmark to identify training needs, develop job profiles/descriptions, develop curricula, and learning programmes in various sectors where the occupation exists. In the Manufacturing sector, demonstration of competence against this NOS may be required in order to run a business or practice a craft or profession.

JUSTIFICATION

A Maintenance Technician is critical in the promotion of safety, longevity, and reliability of equipment in the workplace. His/her role is to oversee, maintain, and ensure overall effectiveness and efficiency of equipment and/or tools under their care. Following the growth in the manufacturing sector, and the heavy investments going into the acquisition of various equipment, it has become imperative that such investments are not wasted due to lack of proper maintenance. Maintenance Technicians play a critical role in the sector by not only ensuring the equipment is effective and efficient through routine and preventive maintenance, but is also safe to operate, and quality output is assured. This National Occupational Standard highlights the core knowledge, skills, competences, and personal attributes that Maintenance Technicians must possess to be successful in their jobs.

ACRONYMS AND ABBREVIATIONS

CS	Core Skills
K	Knowledge and Understanding
MT	Maintenance Technician
NOS	National Occupational Standard
NOSDT	National Occupational Standards Development Team
OK	Organisational Context
OSHE	Occupational Safety, Health and Environment
PC	Performance Criteria
PS	Professional Skills
RK	Regulatory Context
RPL	Recognition of Prior Learning
S	Skills
TK	Technical Knowledge
ZAQA	Zambia Qualifications Authority
ZQF	Zambia Qualifications Framework

GLOSSARY OF TERMS

For the purposes of this NOS, the following terms and definitions shall apply:

Core Skills/Generic Skills: are a group of skills that are key to learning and working in today's world. These skills are typically needed in any work environment. In the context of the NOS, these include communication related skills that are applicable to most job roles.

Disability: is the physical or mental impairment that substantially limits one or more major life activities.

Function: is an activity necessary for achieving the key purpose of the sector, occupation or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of the NOS.

Job Title: defines a unique set of functions that together form a unique employment opportunity in an organisation.

Knowledge and Understanding: are statements which together specify the technical, generic, professional, and organisational specific knowledge that an individual needs in order to perform to the required standard.

National Occupational Standards (NOS): are statements of the standards of performance individuals must achieve when carrying out functions in the workplace, together with specifications of the underpinning knowledge and understanding. They are precise descriptions of what an individual is expected to be able to do in his/her work role.

National Occupational Standards (NOS) Code: is a unique reference code that identifies a NOS.

National Occupational Standards Development Team (NOSDT): means an established group of national stakeholders/experts responsible for the development of National Occupational Standards within a specific economic sector or occupation.

Occupation: is a set of job roles which perform similar/related set of functions in an industry.

Organisational Context: includes the way the organisation is structured and how it operates, including the extent of operative knowledge that managers have in their relevant areas of responsibility.

Performance Criteria: are statements that together specify the standard of performance required when carrying out a task.

Scope: is the set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on the quality of performance required.

Sector: is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy, whose components share similar characteristics and interests.

Sub Sector: is derived from a further breakdown based on the characteristics and interests of its components.

Technical Knowledge: is the specific knowledge needed to accomplish specific designated responsibilities.

Unit Title: gives a clear overall statement about what the incumbent should be able to do.

Elements: set out competences the incumbent should possess to carry out the day-to-day activities.

1. OVERVIEW

This is an introductory section providing a summary and specific information or commentary about the content of the NOS, and the targeted sector and occupation to help the user judge whether it is relevant to them.

NOS Code	NOS.MT.01
Occupation	Maintenance Technician
Job Title	Maintenance Technician
Job Description	A Maintenance Technician is responsible for assembling, installing, and maintaining equipment in the organisation. He/she must be able to read, interpret, and follow instructions and operation manuals; troubleshoot equipment and schedule maintenance activities in the organisation in the form of preventive maintenance.
Job Purpose	Maintenance Technicians work in a variety of industries and ensure equipment is properly assembled, installed, and maintained for optimum output.
ZQF Level	5
Sector	Manufacturing
Sub sectors	Food and Beverage Processing, Petroleum Products Processing, Wood and Wood Products Processing, Metal and Metal Products Processing, Soaps and Detergents Processing, Glass Processing, Fertiliser Manufacturing, Automotive Manufacturing, Consumer Electronics, Chemical Manufacturing, etc.
Other Economic Sector(s) in which the Occupation is Practiced	Mining and Mineral Processing, Aviation/Aerospace, Telecommunication, Water and Sanitation, Government Ministries and Agencies, etc.
Other Similar Jobs that can be Performed in the Occupation	Maintenance Specialist, Service Technician, Industrial Maintenance Technician, Equipment Maintenance Technician, Manufacturing Maintenance Technician, etc.
Minimum Educational Job Entry Qualification(s)	Level 5 Certificate
Practicing License Requirements (if any)	Engineering Institution of Zambia
Training/RPL	1. Awareness of the Industry Standards, and Rules and Regulations and their application. 2. Use of ICTs (Internet, Computer packages, email, Computer Software and Hardware necessary for the job, etc.)

	3. Quality Enhancement Methods 4. Workshop Safety 5. System Maintenance
Minimum Job Entry Age	21 years
Prior Experience (Suggested)	Minimum of 6 months internship
Performance Criteria	As described in the Units under Section 4

2. SCOPE

This NOS highlights the core competences, skills, knowledge and understanding, and personal attributes that Maintenance Technicians are required to possess to be successful in their jobs.

3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES)

This job requires an individual to possess exceptional technical and problem-solving skills. He/she must have the ability to communicate effectively and clearly; plan, prioritise, and reason logically. The jobholder must also be quality conscious, occupational health and safety orientated, have physical fitness, be innovative, computer literate, as well as be able to use fingers, hands, and feet with ease to complete the assigned task (dexterity). He/she must have the ability to handle various tools and materials related to maintenance work. The Individual is also expected to be self-motivated and a great team player.

4. UNITS AND ELEMENTS

This National Occupational Standard is divided into 6 Units, representing the tasks that a jobholder should undertake in his/her day-to-day work. Each unit is further broken down into elements depicting the number of activities to be carried out for the successful execution of a particular task.

UNIT 1 [This Unit covers the skills and knowledge required by a Maintenance Technician for reading and interpreting blueprints; referring and adhering to the instructions provided]

Unit No.	01
Unit Title	Read and Interpret Blueprints, Refer and Adhere to the Instructions Provided
Description	This Unit describes the skills and knowledge required to read and interpret blueprints; referring and adhering to the instructions provided.
Scope	This Unit covers the following: • Reading and Interpreting Blueprints. • Referring and Adhering to the Instructions Provided.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Reading and Interpreting Blueprints	To be competent, the individual must be able to: PC1. Review the title and cover page to get the preliminary information needed to make sense of the rest of the plans. PC2. Analyse the key and legend as this is the explanation for the decoding and understanding of basic symbols in the drawings. PC3. Determine the scale as each unit of measure on a blueprint represents a specific larger unit. The individual must keep in mind that a scale can change from page to page within the same set of prints. PC4. Identify a compass symbol that establishes the drawings' orientation. PC5. Recognise any additional notes as sometimes, specific blueprints include information about the task at hand. PC6. Determine which drawings will be needed first; for example, if one is completing electrical works, they may need to look for the mechanical sections.
Referring and Adhering to the Instructions Provided	To be competent, the individual must be able to: PC7. Interpret information blocks in order to understand: - surface texture and intended usage; - blueprint number; - part number that corresponds to the component; - specific location of the parts and scale information provided. PC8. Identify the material list and components, and equipment needed to properly build each component. PC9. Interpret the key or legend provided. PC10. Interpret the scale at which plans are made, and the symbols and acronyms used. PC11. Analyse the plans as a whole.

Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Standard procedures for reading blueprints and instructions. OK2. Safety rules and regulations for handling equipment and tools in the workshop. OK3. Personal protection, including the use of related safety gear and equipment. OK4. Reporting procedures.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. How to read and interpret blueprints. TK2. Standards and codes of practice applicable to manufacturing TK3. Manufacturers guidelines/specifications for use of equipment and tools. TK4. Basic computer knowledge to update computerised logging systems, as well as work within any company specific programming.
C. Regulatory Context (Knowledge of rules and regulations)	The individual on the job must demonstrate knowledge and understanding of Acts relevant to the manufacturing sector, such as the: RK1. Factories Act. RK2. The National Technical Regulation Act. RK3. Engineering Institution of Zambia Act. RK4. Occupational Health and Safety Act. RK5. Workers' Compensation Act.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English (at least working level) and/or have the means to give simple instructions in the local language used at the workshop. CS2. Prepare and provide clear and simple instructions, details and sketches to co-workers.
	Reading Skills
	The individual on the job must be able to: CS3. Read English, and/or have the means to give simple instructions in the local language used at the workshop. CS4. Read and interpret blueprints or instructions provided for the required work.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS5. Listen attentively and interpret communication/ instructions from the supervisor and co-workers CS6. Convey information clearly and concisely to co-workers.

B. Professional Skills	Decision-Making
	The individual on the job must be able to:
	PS1. Determine and report to the superior the presence of possible errors in the blueprints.
	PS2. Determine and report to the superior the need to customise instructions if required.
	Plan and Organise
	The individual on the job should be able to:
	PS3. Plan work and organise required resources in coordination with team members and superiors.
	PS4. Prepare a safe working environment for maintenance works.
	PS5. Schedule daily works as per the maintenance requirements.
	Customer Centricity
	The individual on the job should be able to:
	PS6. Support the organisation's commitment to customer expectation of quality output by ensuring adherence to maintenance requirements.
	Problem-Solving
	The individual on the job should be able to:
	PS7. Resolve any conflicts within the team.
	PS8. Propose changes to increase efficiency of the equipment
	Analytical Thinking
	The individual on the job should be able to:
	PS9. Analyse and convey to the superior any equipment malfunction and carry out remedial action.
	PS10. Evaluate the state of the equipment.
	PS11. Recognise trends in equipment failure or malfunction.
	Critical Thinking
	The individual on the job should be able to:
	PS12. Identify and deal with or report violation of any safety norms which may lead to accidents.
	PS13. Justify a maintenance schedule that is customised to the specific conditions of the equipment and environment in which it operates.
	PS14. Question information that is presented without evidence and be willing to change their viewpoint when presented with new evidence.

UNIT 2 [This Unit covers the skills and knowledge required by a Maintenance Technician for troubleshooting and diagnosing equipment problems]

Unit No.	02
Unit Title	Troubleshoot and Diagnose Equipment Problems
Description	This Unit describes the skills and knowledge required to troubleshoot and diagnose equipment problems.
Scope	This Unit covers the following: • Troubleshoot Equipment Problems. • Diagnose Equipment Problems.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Troubleshoot Equipment Problems	To be competent, the individual must be able to: PC1. Identify the problem. PC2. Examine a theory of probable cause. PC3. Test the theory to determine the cause. PC4. Formulate a plan of action to resolve the problem. PC5. Implement the solution. PC6. Evaluate full system functionality. PC7. Propose preventive measures.
Diagnose Equipment Problems	To be competent, the individual must be able to: PC8. Analyse equipment operating conditions in order to produce a list of possible failure modes such as: - performance fluctuations; - error codes; - detection of leakages, etc. PC9. Use subjective evidence (analogy) based on previous experience to further isolate the specific cause of a problem using: - sight; - sound; - smell; - repair history; - adjacent system data. PC10. Apply fault estimation techniques to determine the magnitude and severity of the fault.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Standard procedures for troubleshooting equipment problems. OK2. Safety rules and regulations for handling equipment and tools in the workshop. OK3. Personal protection, including the use of related safety gear and equipment. OK4. Reporting procedures.

B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Troubleshooting techniques and procedures. TK2. Standards and codes of practice applicable to manufacturing. TK3. Manufacturers guidelines/specifications for use of equipment and tools. TK4. Basic computer knowledge to update computerised logging systems, as well as work within any company specific programming. TK5. Information gathering. TK6. Assessment of the effectiveness of a proposed solution. TK7. Documentation of the incident.
C. Regulatory Context (Knowledge of rules and regulations)	The individual on the job must demonstrate knowledge and understanding of Acts relevant to the manufacturing sector, such as the: RK1. Factories Act. RK2. National Technical Regulation Act. RK3. Engineering Institution of Zambia Act. RK4. Occupational Health and Safety Act. RK5. Workers' Compensation Act.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English (at least working level) and/or have the means to give simple instructions in the local language used at the workshop. CS2. Prepare and provide clear and simple instructions, details and sketches to co-workers.
	Reading Skills
	The individual on the job must be able to: CS3. Read English, and/or have the means to give simple instructions in the local language used at the workshop CS4. Read and interpret instructions provided for the required work.
B. Professional Skills	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS5. Listen attentively and interpret communication/ instructions from the supervisor and co-workers. CS6. Convey information clearly and concisely to co-workers.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Determine and report to the superior the presence of possible errors in the equipment. PS2. Determine and report to the superior the need to isolate affected components.

	Plan and Organise
	The individual on the job should be able to:
	PS3. Plan work and organise required resources in coordination with team members and superiors.
	PS4. Prepare a safe working environment for the maintenance works.
	PS5. Schedule daily works as per the maintenance requirements.
	Customer Centricity
	The individual on the job should be able to:
	PS6. Support the organisation's commitment to customer expectation of quality output by ensuring adherence to maintenance requirements.
	Problem-Solving
	The individual on the job should be able to:
	PS7. Resolve any conflicts within the team.
	PS8. Estimate the magnitude and severity of the fault.
	PS9. Identify required tools to fix identified fault.
	PS10. Propose changes to increase efficiency of the equipment
	Analytical Thinking
	The individual on the job should be able to:
	PS11. Evaluate the state of the equipment.
	PS12. Analyse and convey to the superior any equipment malfunction and carry out remedial action.
	PS13. Recognise trends in equipment failure or malfunction.
	Critical Thinking
	The individual on the job should be able to:
	PS14. Identify and deal with or report violation of any safety norms which may lead to accidents.
	PS15. Question information that is presented without evidence, and be willing to change their viewpoint when presented with new evidence.

UNIT 3 [This unit covers the skills and knowledge required by a Maintenance Technician for repairing or replacing broken or malfunctioning parts and conducting tests to ensure the repairs are satisfactorily done]

Unit No.	03
Unit Title	Repair or Replace Broken or Malfunctioning Parts and Conduct Tests
Description	This Unit describes the skills and knowledge required to repair or replace broken or malfunctioning parts, and conducting tests to ensure the repairs are done in a satisfactory manner.
Scope	This Unit covers the following: - Repair or Replacement of Broken or Malfunctioning Parts - Conduct Tests on Repaired Equipment.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Repair or Replacement of Broken or Malfunctioning Parts	To be competent, the individual must be able to: PC1. Identify the failure (either an equipment component or equipment item). PC2. Diagnose the problem. PC3. Eliminate the part causing the failure. PC4. Recommend repair or replacement when failure occurs, and follow troubleshooting procedures. PC5. Order the replacement (internally or externally) of the part should the decision be a replacement, or else repair the failed part. PC6. Install ordered or repaired part.
Conduct Tests on Repaired Equipment	To be competent, the individual must be able to: PC7. Test functionality of the system after repair or replacement of affected part. PC8. Assess continuation of use in the system.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Standard procedures for conducting maintenance procedures. OK2. Safety rules and regulations for handling equipment and tools in the workshop. OK3. Personal protection, including the use of related safety gear and equipment. OK4. Reporting procedures.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Installation of various equipment relevant to the organisation. TK2. Standards and codes of practice applicable to maintenance in the manufacturing sector. TK3. Maintenance guidelines/specifications for use of equipment and tools.

	TK4. Basic computer knowledge to update computerised logging systems, as well as work within any company specific programming. TK5. Repairing techniques relevant to the equipment in the organisation.
C. Regulatory Context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of Acts relevant to the Manufacturing sector, such as the: RK1. Factories Act. RK2. National Technical Regulation Act. RK3. Engineering Institution of Zambia Act. RK4. Occupational Health and Safety Act. RK5. Workers' Compensation Act.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English, (at least working level) and/or have the means to give simple instructions in the local language used at the workshop. CS2. Prepare and provide clear and simple instructions, details and sketches to co-workers.
	Reading Skills
	The individual on the job must be able to: CS3. Read English, and/or have the means to give simple instructions in the local language used at the workshop. CS4. Read and interpret maintenance instructions provided for the required work.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS5. Listen attentively and interpret communication/ instructions from the supervisor and co-workers. CS6. Convey information clearly and concisely to co-workers.
B. Professional Skills	Decision-Making
	The individual on the job must be able to: PS1. Determine and report to the superior the need to eliminate malfunctioning parts. PS2. Determine and report to the superior repair or replacement of the malfunctioning parts.
	Plan and Organise
	The individual on the job should be able to: PS3. Plan work and organise required resources in coordination with team members and superiors. PS4. Prepare a safe working environment for the maintenance work. PS5. Schedule daily works as per the maintenance requirements.

	Customer Centricity
	The individual on the job should be able to:
	PS6. Support the organisation's commitment to customer expectation of quality output by ensuring adherence to maintenance requirements.
	Problem-Solving
	The individual on the job should be able to:
	PS7. Resolve any conflicts within the team.
	PS8. Estimate the magnitude and severity of the fault.
	PS9. Identify required tools to fix identified fault.
	PS10. Propose changes to increase efficiency of the equipment
	PS11. Identify potential safety hazards to avoid injuries.
	Analytical Thinking
	The individual on the job should be able to:
	PS12. Analyse and convey to the superior any equipment malfunction and carry out remedial action.
	PS13. Evaluate the state of the equipment.
	PS14. Recognise trends in equipment failure or malfunction.
	Critical Thinking
	The individual on the job should be able to:
	PS15. Identify and deal with or report violation of any safety norms which may lead to accidents.
	PS16. Justify a maintenance schedule that is customised to the specific conditions of the equipment and environment in which it operates.
	PS17. Question information that is presented without evidence, and be willing to change their viewpoint when presented with new evidence.

UNIT 4 [This Unit covers the skills and knowledge required by a Maintenance Technician to perform equipment installations and adjustments]

Unit No.	04
Unit Title	Perform Equipment Installations and Adjustments
Description	This Unit describes the skills and knowledge required to perform equipment installations and adjustments.
Scope	This Unit covers the following: • Perform Equipment Installations. • Perform Equipment Adjustments.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Perform Equipment Installations	To be competent, the individual must be able to: PC1. Plan and prepare for equipment installation, which includes site evaluation, assessment of equipment needs, and obtaining necessary permits and approvals. PC2. Create a detailed installation plan outlining the steps to follow during the installation process. PC3. Arrange equipment delivery, unloading, and inspection for damages or defects. PC4. Organise when the equipment will be installed and who would be responsible for the installation. PC5. Schedule the duration of the installation process. PC6. Assemble the equipment according to the installation plan. PC7. Test the equipment to ensure proper functionality. PC8. Conclude the installation process and recommend commissioning of the equipment to superiors.
Perform Equipment Adjustments	To be competent, the individual must be able to: PC9. Test equipment after installation to ensure proper functionality. PC10. Identify and resolve performance issues that arise during testing. PC11. Propose reconfiguration to accommodate changes in equipment already running. PC12. Perform adjustments in accordance with the manufacturer's guidelines during maintenance to prolong lifespan of the equipment.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Standard procedures for conducting maintenance. OK2. Safety rules and regulations for handling equipment and tools in the workshop. OK3. Personal protection, including the use of related safety gear and equipment OK4. Reporting procedures.

B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of local manufacturing codes and mandatory standards below: TK1. Installation of various equipment relevant to the organisation. TK2. Standards and codes of practice applicable to maintenance in the manufacturing sector. TK3. Installation and maintenance guidelines/specifications for use of equipment and tools. TK4. Basic computer knowledge to update computerised logging systems, as well as work within any company specific programming. TK5. Repairing techniques relevant to the equipment in the organisation.
C. Regulatory Context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of Acts relevant to the manufacturing sector, such as the: RK1. Factories Act. RK2. National Technical Regulation Act. RK3. Engineering Institution of Zambia Act. RK4. Occupational Health and Safety Act. RK5. Workers' Compensation Act.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English (at least working level) and/or have the means to give simple instructions in the local language used at the workshop. CS2. Prepare and provide clear and simple instructions, details and sketches to co-workers.
	Reading Skills
	The individual on the job must be able to: CS3. Read English, and/or have the means to give simple instructions in the local language used at the workshop. CS4. Read and interpret installation and maintenance instructions provided for the required work.
	Oral Communication (Listening and Speaking Skills)
B. Professional Skills	The individual on the job must be able to: CS5. Listen attentively and interpret communication/ instructions from the supervisor and other co-workers. CS6. Convey information clearly and concisely to co-workers.
	Decision-Making
	The individual on the job must be able to: PS1. Determine and report to the superior the need to commission installed equipment. PS2. Determine and report to the superior the adjustment decisions on installed parts.

	PS3. Determine and report to the superior the reconfiguration decision on installed and running equipment.
	Plan and Organise
	The individual on the job should be able to:
	PS4. Plan work and organise required resources in coordination with team members and superiors.
	PS5. Prepare a safe working environment for the maintenance works.
	PS6. Schedule daily works as per the maintenance requirements.
	Customer Centricity
	The individual on the job should be able to:
	PS7. Support the organisation's commitment to customer expectation of quality output by ensuring adherence to maintenance requirements.
	Problem-Solving
	The individual on the job should be able to:
	PS8. Resolve any conflicts within the team.
	PS9. Justify suitability of the job site.
	PS10. Identify damage or defects by inspecting equipment due for installation.
	PS11. Propose changes to increase efficiency of the equipment
	PS12. Identify potential safety hazards to avoid injuries.
	Analytical Thinking
	The individual on the job should be able to:
	PS13. Analyse and convey to the superior any equipment malfunction and carry out remedial action.
	PS14. Evaluate the state of the equipment.
	PS15. Recognise trends in equipment failure or malfunction.
	Critical Thinking
	The individual on the job should be able to:
	PS16. Identify and deal with or report violation of any safety norms which may lead to accidents.
	PS17. Justify a maintenance schedule that is customised to the specific conditions of the equipment and environment in which it operates.
	PS18. Question information that is presented without evidence, and be willing to change their viewpoint when presented with new evidence.

UNIT 5 [This Unit covers the skills and knowledge required by a Maintenance Technician in monitoring performance of machinery and equipment]

Unit No.	05
Unit Title	Monitor Performance of Machinery and Equipment
Description	This Unit describes the skills and knowledge required to monitor the performance of machinery and equipment.
Scope	This Unit covers the following: • Monitor Performance of Machinery and Equipment
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Monitor Performance of Machinery and Equipment	To be competent, the individual must be able to: PC1. Interpret different alarms and their status variations. PC2. Operate various monitoring and measuring devices Simultaneously. PC3. Differentiate monitoring techniques depending on affected equipment or system type. PC4. Assess the condition of various equipment simultaneously. PC5. Compare data of an entire manufacturing process.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Monitoring techniques. OK2. Safety rules and regulations for handling equipment and tools in the workshop. OK3. Personal protection, including the use of related safety gear and equipment. OK4. Reporting procedures.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of local manufacturing codes and mandatory standards below: TK1. Interpretation of various alarms. TK2. Standards and codes of practice applicable to the manufacturing sector. TK3. Basic computer knowledge to update computerised logging systems, as well as work within any company specific programming. TK4. Monitoring techniques relevant to the equipment in the organisation.
C. Regulatory Context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of Acts relevant to the manufacturing sector, such as the: RK1. Factories Act. RK2. National Technical Regulation Act. RK3. Engineering Institution of Zambia Act. RK4. Occupational Health and Safety Act. RK5. Workers' Compensation Act.

Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English (at least working level) and be able to or have the means to give simple instructions in the local language used at the workshop. CS2. Prepare and provide clear and simple instructions, details and sketches to co-workers.
	Reading Skills
	The individual on the job must be able to: CS3. Read English, and be able to or have the means to give simple instructions in the local language used at the workshop. CS4. Read and interpret monitoring instructions provided for the required work.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS5. Listen attentively and interpret communication/ instructions from the supervisor and co-workers. CS6. Convey information clearly and concisely to co-workers.
B. Professional Skills	Decision-Making
	The individual on the job must be able to: PS1. Determine and report to the superior on persistent alarms. PS2. Determine and report to the superior the adjustment decisions on alarm calibrations.
	Plan and Organise
	The individual on the job should be able to: PS3. Plan work and organise required resources in coordination with team members and superiors. PS4. Prepare a safe working environment for the maintenance works. PS5. Schedule daily works as per the monitoring requirements
	Customer Centricity
	The individual on the job should be able to: PS6. Support the organisation's commitment to customer expectation of quality output by ensuring adherence to monitoring requirements.
	Problem-Solving
	The individual on the job should be able to: PS7. Resolve any conflicts within the team. PS8. Identify defective monitoring equipment. PS9. Propose changes to increase efficiency of the equipment. PS10. Identify potential safety hazards to avoid injuries.

	Analytical Thinking
	The individual on the job should be able to:
	PS11. Analyse and convey to the superior any equipment malfunction and carry out remedial action.
	PS12. Evaluate the state of monitoring equipment.
	PS13. Recognise trends in equipment failure or malfunction.
	Critical Thinking
	The individual on the job should be able to:
	PS14. Identify and deal with or report violation of any safety norms which may lead to accidents.
	PS15. Justify a maintenance schedule that is customised to the specific conditions of the equipment and environment in which it operates.

UNIT 6 [This Unit covers the skills and knowledge required by a Maintenance Technician in maintaining a preventive maintenance schedule for inspecting, servicing, and testing equipment]

Unit No.	06
Unit Title	Maintain a Preventive Maintenance Schedule for Inspecting, Servicing, and Testing Equipment
Description	This Unit describes the skills and knowledge required to maintain a preventive maintenance schedule for inspecting, servicing, and testing equipment.
Scope	This Unit covers the following: Preventive Maintenance Scheduling for Inspection, Servicing, and Testing Equipment.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Preventive Maintenance Scheduling for Inspection, Servicing, and Testing Equipment	To be competent, the individual must be able to: PC1. Create an inventory of the organisation's assets. PC2. Arrange maintenance priorities. PC3. Identify appropriate maintenance intervals. PC4. Schedule recurring inspections and tasks. PC5. Evaluate progress and keep fine-tuning.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Standard procedures for conducting preventive maintenance. OK2. Safety rules and regulations for handling equipment and tools in the workshop. OK3. Personal protection, including the use of related safety gear and equipment. OK4. Reporting procedures.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of local manufacturing codes, mandatory standards, and skills below: TK1. Standards and codes of practice applicable to preventive maintenance in the manufacturing sector. TK2. Installation and maintenance guidelines/specifications for use of equipment and tools. TK3. Basic computer knowledge to update computerised logging systems, as well as work within any company specific programming. TK4. Basic inventory management knowledge. TK5. Preventive maintenance techniques relevant to the equipment in the organisation.
C. Regulatory Context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of Acts relevant to the manufacturing sector, such as the: RK1. Factories Act.

	RK2. National Technical Regulation Act. RK3. Engineering Institution of Zambia Act. RK4. Occupational Health and Safety Act. RK5. Workers' Compensation Act.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills The individual on the job must be able to: CS1. Write in English (at least working level) and/or have the means to give simple instructions in the local language used at the workshop. CS2. Prepare and provide clear and simple instructions, details and sketches to co-workers.
	Reading Skills The individual on the job must be able to: CS3. Read English, and/or have the means to give simple instructions in the local language used at the workshop CS4. Read and interpret installation and maintenance instructions provided for the required work.
	Oral Communication (Listening and Speaking skills) The individual on the job must be able to: CS5. Listen attentively and interpret communication/ instructions from the supervisor and co-workers. CS6. Convey information clearly and concisely to co-workers.
	Decision-Making The individual on the job must be able to: PS1. Determine and report to the superior prioritised preventive maintenance schedules. PS2. Determine and report to the superior the adjustment decisions on installed parts. PS3. Determine and report to the superior the reconfiguration decisions on installed and running equipment.
	Plan and Organise The individual on the job should be able to: PS4. Plan work and organise required resources in coordination with team members and superiors. PS5. Prepare a safe working environment for the maintenance works. PS6. Schedule daily works as per the maintenance requirements.
B. Professional Skills	Customer Centricity The individual on the job should be able to: PS7. Support the organisation's commitment to customer expectation of quality output by ensuring adherence to maintenance requirements.

	Problem-Solving
	The individual on the job should be able to:
	PS8. Resolve any conflicts within the team.
	PS9. Change damaged or defective components or parts identified from equipment inspections.
	PS10. Identify potential safety hazards to avoid injuries.
	Analytical Thinking
	The individual on the job should be able to:
	PS11. Analyse and convey to the superior any equipment malfunction and carry out remedial action.
	PS12. Evaluate the state of the equipment.
	PS13. Recognise trends in equipment failure or malfunction.
	Critical Thinking
	The individual on the job should be able to:
	PS14. Identify and deal with or report violation of any safety norms which may lead to accidents.
	PS15. Justify a maintenance schedule that is customised to the specific conditions of the equipment and environment in which it operates.

5. EQUIPMENT, TOOLS AND CONSUMABLE MATERIALS

These include, but not limited to:

Equipment and Tools:

- Computer software applications
- Machine shop equipment and tools
- Fabrication shop equipment and tools
- Electrical equipment
- Testing equipment and tools
- Personal protective equipment
- Lifting equipment
- Climbing ladders
- Safety warning and information signs
- Incident/accident reporting templates
- Company standard operating procedures
- First Aid kit
- Safety tools and equipment such as fire extinguishers and barricades
- Company's safety policy/procedure
- Zambia Standards and recognised codes of practice applicable to the Manufacturing sector

Raw Materials and Consumables: Textile, leather, wood, paper, chemicals, rubber, plastics, non-metallic minerals, basic metals, lubricants, spare parts, and components, etc.

6. JOBHOLDER DILEMMAS/CHALLENGES AND COMPLEXITIES

Maintenance Technicians face challenges such as obsolete and/or inappropriate equipment and tools, budgetary constraints, inadequate product costing skills, poor technical skill base, bureaucracy in procurement procedures, lack of appreciation of preventive maintenance by non-technical management staff, labour intensive nature of the work, rapid change of technology and materials, lack of personal protective equipment, and inconsistency in company and government policies and regulations, etc.

6.1 Alternative Choices (Solutions) to Dilemmas and Complexities

Solutions to challenges include: selecting and procuring appropriate equipment and tools for the job; supporting capacity building through training; recruitment or inclusion of engineering professionals in management teams; deployment of automation where feasible; provision of personal protective equipment; participation in lobbying and formulation of policies; allocation of adequate financial resources, etc.

7. WORKING CONDITIONS/ENVIRONMENT

Maintenance Technicians work with a variety of machinery, toxic substances, and volatile materials. Their work environment is susceptible to fires, explosions, structural failures, and equipment malfunctions. Working conditions include: cold, hot, and wet conditions, climbing heights, standing and/or walking for long hours, lifting materials, working in day or night shifts, working in areas that are noisy and dusty, as well as areas with limited lighting and ventilation, etc.

8. PARTIES INVOLVED/INTERACTING WITH THE JOB HOLDER OR TRAINEE

8.1 Internal/Within the Organisation

Management, supervisors, subordinates, and other section members, etc.

8.2 External/Outside the Organisation

Government regulators, professional bodies, suppliers, fellow technicians from other companies, labour unions, students/interns, etc.

9. DEMANDS ON THE PHYSICAL BODY

- Physique to sustain strenuous conditions;
- Be able to walk and stand for long periods of time;
- Bend, stretch, twist, or reach out;
- Be able to lift relatively heavy materials, tools and equipment;
- Be able to use fingers, hands, and feet with ease to complete the assigned task (dexterity);

ANNEX A

Criteria for Assessments based on this NOS

A.1 Guidelines for Assessment

A.1.1 Criteria for assessment for curricula and learning programmes based on this NOS will be created by curricula and programme developers. Each Performance Criteria (PC) will be assigned marks proportional to its importance in the NOS. Curricula and programme developers will also lay down proportion of marks for theory and practical skills for each performance criteria, giving more weight to practical skills.

There shall be allocated the 'Total Mark', which will be the sum of all marks in each Unit, distributed across the number of PCs in that particular Unit. The 'Out of' mark will be the mark allocated to each PC, which will be shared between theory and skills practical assessments.

A.1.2 Individual awarding/assessment bodies or institutions and other users of the NOS will create unique question papers for the theory part and evaluations for skill practical part for their respective candidates.

ANNEX B

NOS Version Control

This Annex gives details necessary for the tracking of the NOS versions based on the number of revisions.

NOS Code	NOS.MT.01		
ZQF Level	5	Version Number	01
Sector	Manufacturing	Date of Approval	February 2024
Sub Sector	Food and Beverage Processing, Petroleum Products Processing, Wood and Wood Products Processing, Metal and Metal Products Processing, Soaps and Detergents Processing, Glass Processing, Fertiliser Manufacturing, Automotive Manufacturing, Consumer Electronics, Chemical Manufacturing, etc.	Date of Last Review	N/A
Occupation	Maintenance Technician	Date of Next Review	February 2029



*“You learn,
We Standardise”*

ZAMBIA QUALIFICATIONS AUTHORITY

Finsbury Park, Ground Floor, Kabwe Roundabout
P.O Box 51103, Lusaka, Zambia

+260 211 843 050 // +260 963 922 730
+260 956 037 185 // +260 972 559 301

info@zaqa.gov.zm
www.zaqa.gov.zm

<https://www.linkedin.com/company/zambia-qualifications-authority/>
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